[INCH-POUND]
A-A-59250
June 16, 1998
SUPERSEDING
KKK-E-2808
December 30, 1992

COMMERCIAL ITEM DESCRIPTION

EXCAVATOR, MULTIPURPOSE, TRUCK MOUNTED, TWO CABS

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers a truck mounted, two cab, two diesel-engine, telescoping boom type multipurpose excavator.

2. CLASSIFICATION. The excavator shall be one of the following sizes and classes, as specified (see 7.2):

Size 1A - (Excavator without outriggers)
Lifting capacity (with bucket) not less than 5,100 pounds (lb) at 15-foot (ft) (2 313 kilogram (kg) at 4.57 metre (m)) radius, bucket at ground level, measured in accordance with SAE J 1097.

Size 1B - (Excavator with outriggers)
Lifting capacity (with bucket) not less than 6,700 lb at 15-foot (3 038 kg at 4.57 m) radius, bucket at ground level, measured in accordance with SAE J 1097.

Size 2A - (Excavator without outriggers)
Lifting capacity (with bucket) not less than 6,650 lb at 15-ft (3 016 kg at 4.57 m) radius, bucket at ground level, measured in accordance with SAE J 1097.

Size 2B - (Excavator with outriggers)
Lifting capacity (with bucket) not less than 9,100 lb at 15-foot (4 127 kg at 4.57 m) radius, bucket at ground level, measured in accordance with SAE J 1097.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3805

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Class B - 6 by 4 (3 axles, rear tandem axle driven).

Class C - 6 by 6 (3 axles, rear tandem, all wheel drive).

3. SALIENT CHARACTERISTICS.

3.1 Description. The excavator shall consist of a revolving upperstructure, with a bucket, ripper, or grading blade attached on the end of a telescoping boom. The assembly is mounted on a pneumatic-tired truck type carrier. The revolving upperstructure shall consist of operator's cab, all operational controls, hydraulic system, and the telescopic boom. The diesel engine power unit, through the hydraulic system, shall provide motivation for all functions, except travel. The travel function is motivated by a separate diesel engine and mounted on the carrier.

3.2 Standard commercial product. The excavator shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this commercial item description but which are a part of the manufacturer's standard commercial product, shall be included in the excavators being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model. Additional options may be added at time of issuance of contract or purchase order as required by purchasing agency.

3.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Safety. Rotating or reciprocating parts and parts subject to high operational temperatures, that are of a nature or are located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or guarded. The principal walking platform surfaces shall be of an anti-skid type. Hand holds and steps shall be provided to allow safe entrance and exit from the operator's cab and the truck cab. The excavator shall meet current safety standards of Occupational Safety and Health Administration (OSHA) and Power Crane and Shovel Association (PCSA) guidelines.

3.4.1 Noise. When the noise level, tested in accordance with MIL-STD-1474, is greater than 85 decibels (db(A)), a caution plate shall be permanently attached in a conspicuous location clearly visible to all operating personnel. The caution plate shall read: CAUTION - HEARING PROTECTION REQUIRED. The plate shall have a yellow background with black lettering and plate shall be made of corrosion-resistant material. The caution plate shall be securely affixed to the equipment with nonferrous screws or bolts.

3.4.2 Electrical warning decal. An electrical warning decal shall be provided, in accordance with OSHA requirements, cautioning the operator or driver about operating the excavator near electric power lines.

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3.4.3 Strike hazard. The back end of the revolving upperstructure, and the telescoping portion of the boom, shall be painted with black stripes to denote a hazard to personnel on the ground.

3.4.4 Backup alarm. The excavator shall be furnished with a reverse alarm signal meeting requirements of CFR, paragraph 1926.601.

3.5 Environmental conditions. The excavator shall start and operate as specified herein under any of the following conditions or combination of conditions:

a. In any ambient temperature from - 20 degrees Fahrenheit (°F) to 110 °F (- 29 degrees Celsius (°C) to 43 °C), except as specified in 3.5b below.

b. At any elevation from sea level (barometric pressure 29.92 inches in mercury (101.3 kilopascal (kPa)) and a maximum ambient temperature of 110 °F (43 °C) to 5,000-ft (1 524 m) (barometric pressure 24.89 inches mercury (84.27 kPa)) and a temperature of 100 °F (38 °C).

3.6 Revolving upperstructure.

3.6.1 Cab. The cab shall be fabricated of sheet metal or sheet fiberglass and shall have sufficient windows and glazed doors to permit 270 degrees of vision for the operator. All glass shall be of the safety type. The front window shall be arranged for easy opening or removal. Provision shall be made to fold or store the front window clear of the operator. Cab doors shall be adequately restrained from accidentally opening or closing while the excavator is traveling or operating. A comfortable seat for the operator with four-way adjustment shall be provided. The cab shall be provided with a heater and defroster. When specified (see 7.2), the cab shall be provided with controls for operating the carrier from the cab of the upperstructure. These controls shall permit the excavator operator to operate the carrier's brakes, steering, and the forward and reverse directions. These controls shall not interfere with the operation of the conventional carrier controls.

3.6.2 Swing stop device. A means of locking the upperstructure in any desired position shall be provided. The swing stop device control shall be in accordance with PCSA Standard No. 5, and shall be located within easy reach of the operator.

3.7 Operating controls. All controls shall be in accordance with SAE J 1177.

3.8 Hydraulic system and operation. All high pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure. High pressure hydraulic hoses conform to SAE J 517. The following functions shall be hydraulically operated and controlled:

a. Telescoping boom.
b. Rotation of bucket or boom.

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c. Vertical control of boom.
d. Revolving of upperstructure.

3.8.1 Hydraulic monitoring system. The monitoring system shall alert the operator, either by audio or visual lights, of problems which may damage hydraulic system components. The monitoring system shall also alert the operator of excessively high oil temperatures.

3.9 Boom and attachments.

3.9.1 Boom. The excavator shall be provided with a telescoping boom with the following characteristics and capabilities, when measured in accordance with SAE J 1193:

a. Extended length (AU) not less than 21-ft (6.4 m) for size 1A and size 1B excavators, and not less than 24-ft (7.3 m) for size 2A and 2B excavators.
b. Retracted length (AV) not greater than 13-ft (3.96 m) for size 1A and size 1B excavators, and not greater than 15.75-ft (4.8 m) for size 2A and 2B excavators.
c. Maximum digging depth (AB) not less than 15-ft (4.57 m) (below ground level) for all sizes of excavators.
d. Digging reach (AA) not less than 26-ft (7.9 m) for size 1A and size 1B excavators, and not less than 29-ft (8.8 m) for size 2A and size 2B excavators.
e. Dumping height (BD) not less than 15-ft (4.57 m) for all sizes of excavator.
f. Digging through a range around the excavator not less than 270 degrees.
g. Swing shall be continuous.
h. Boom or bucket rotation or tilt (AX) not less than 45 degrees to right and left.

3.9.2 Attachments. The following excavating attachments and accessories shall be furnished as specified (see 7.2):

	Minimum capacity (struck) cubic yard (yd^3)	cubic metre (m^3)
a. Size 1A and size 1B excavators:		
Bucket, excavating, 24-inch (610 millimetre (mm)) cutting width	0.375	(0.29)
Bucket, ditch cleaning, 60-inch (1 524 mm) width	0.500	(0.38)
Bucket, excavating, 30-inch (762 mm) width	0.500	(0.38)
Bucket, sidewalk removal, 30-inch (762 mm) width		
Bucket, 8-ft (2.4 m) width, with straight edge		
Boom extension, not less than 4-ft (1.2 m), complete with necessary control gear		
b. Size 2A and size 2B excavators:		
Bucket, excavating, 24-inch (610 mm) cutting width	0.375	(0.29)
Bucket, excavating, 36-inch (914 mm) cutting width	0.500	(0.38)
Bucket, dredging, 72-inch (1 829 mm)	0.750	(0.57)
Bucket, ditching, cleaning 60-inch (1 524 mm)	0.625	(0.48)
Bucket, concrete pouring	0.500	(0.38)

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Bucket, pavement removal, not less than
40-inch (1 016 mm)
Ripper, single tooth, heavy duty
Blade, grading 8-ft (2.4 m) width, with straight edge
Boom extension, not less than 4-ft (1.2 m), complete
with necessary control gear
Boom extension, not less than 6-ft (1.8 m), complete
with necessary control gear

The above listed attachments shall fit and operate on the excavator and shall be replaceable without the use of special tools.

3.10 Lifting capacity and tipping load. Loads shall be not greater than 87 percent of hydraulic lifting capacity or 75 percent of tipping capacity. A load chart for the entry digging range shall be provided.

3.11 Carrier. The carrier shall be provided with fenders for the front wheels. Unless otherwise specified (see 7.2), wheels on all axles shall be disk type. The carrier cab shall be provided with a heater and defroster.

3.11.1 Drive axles. As a minimum, one rear drive axle assembly shall be furnished with an increased traction type differential.

3.11.2 Speed. The excavator shall have not less than five forward speeds and one reverse speed, and shall be capable of traveling on level pavement at a maximum speed in high gear of not less than 40 miles per hour (64 kilometre per hour).

3.11.3 Gradeability. The excavator shall be capable of negotiating a one percent grade in the fourth gear and 30 percent grade in the lowest gear.

3.11.4 Boom rack. A suitable boom rack shall be provided for stabilizing the boom when in the travel position. The boom shall be rigidly secured to prevent swinging and striking the carrier cab. Horizontal but not vertical oscillation shall be held to a minimum.

3.12 Outriggers. The outriggers shall be hydraulically powered, locally operated or operated from the revolving upperstructure cab, and shall be capable of being set or retracted independently or simultaneously. Outriggers shall be capable of being adjusted at not less than 4 inches (102 mm) below ground level and 3 inches (76 mm) above ground level to compensate for uneven terrain. A device to accurately determine when the excavator is level shall be provided.

3.13 Brakes. Service and parking brakes shall be in accordance with SAE J 1152.

3.14 Steering. The carrier shall be equipped with power steering.

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3.15 Lighting. The excavator shall be equipped with two recessed sealed beam headlights, two dual stop and taillights, instrument panel light, and directional turn signals. When specified (see 7.2), dual floodlights shall be provided. All lighting shall be in accordance with applicable DoT Federal Motor Carrier Safety Regulations.

3.16 Engines. The engines shall be equipped as a minimum with the following:

a. Hour meter.
b. Tachometer.
c. Cooling liquid temperature indicator.
d. Lubricating oil pressure indicator.
e. Liquid fuel gage.
f. Fuel tank with sufficient capacity for 8 hours normal operation.

3.16.1 Cold starting system. Each excavator diesel engine shall be equipped with an engine priming system or glow plug type for cold temperature engine starting. When an ether priming system is furnished, it shall be of the measured shot type with a reservoir of not less than 12 fluid ounces (355 millilitre), and shall be furnished for each engine.

3.17 Tires. Tires shall be of the tube or tubeless type. The individual rated load-carrying capacities shall be in accordance with TRA recommendations. Tires shall have nondirectional traction type tread.

3.18 Toolbox. The excavator shall be equipped with a lockable metallic toolbox. The toolbox shall be securely fastened to the excavator in a readily accessible and protected location.

3.19 Fire extinguisher. When specified (see 7.2), a fire extinguisher shall be provided. The type and size shall be as specified (see 7.2).

3.20 Lifting and tiedown attachments. When specified (see 7.2), the excavator shall be equipped with lifting and tiedown attachments that meet requirements of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the excavator. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the excavator on the carrier when shipped.

3.21 Instruction plates. The excavator shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous bolts, screws, or rivets. Heavy-duty decals made of material which will last abd remain legible for the life of the equipment are acceptable.

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3.22 Identification plates. An identification plate will be furnished by the contracting officer for each excavator. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each excavator in a conspicuous place with nonferrous screws, rivets, or bolts. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.23 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 7.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to the manufacturer's current process and the total dry film thickness shall be not less than 2.5 mils (0.0635 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.24 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J 534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch (6 895 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.25 Servicing and adjusting. Prior to acceptance of the excavator by the Government, the contractor shall service and adjust the excavator for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment (when required).
e. Adjustment of engine to include tune-up (when required).
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

3.26 Rail transport. When specified (see 7.2), the excavator shall pass the MIL-STD-810 rail impact test without damage or permanent deformation.

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3.27 Air transportability configuration. When specified (see 7.2), the excavator shall be transportable by C-130, C-141, C-5, and C-17 aircraft in a roll-on/roll-off configuration. Refer to MIL-STD-1791, for dimensional and weight guidance for this requirement.

3.28 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of IEEE SI-10 and all other requirements of this CID including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS.

4.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Copies of military standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

7.1.2 The Federal Acquisition Regulation (FAR), DOT Federal Motor Carrier Safety Regulations and OSHA standards may be obtained from the Superintendent of Documents, U.S. Printing Office, Washington, DC 20402.

7.1.3 IEEE Standard is available from the Institute of Electrical and Electronic Engineers, IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.1.4 PCSA Standard is available from the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee, WI 53202.

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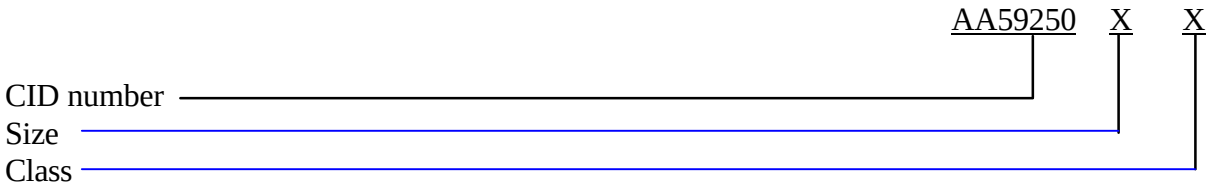
7.1.5 SAE Standards are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.1.6 TRA Standards are available from the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Size and class of excavator required (see 2.).
c. When upperstructure cab is to be provided with controls for carrier (see 3.6.1).
d. List of attachments and accessories required for each size of excavator (see 3.9.2).
e. When other than disk wheels are required for carrier (see 3.11).
f. When dual floodlights are required (see 3.15).
g. When a fire extinguisher is required and the type thereof (see 3.19).
h. When lifting and tiedown attachments are required (see 3.20).
i. Color of finish coat (see 3.23).
j. When rail impact test is required (see 3.26).
k. When air transportability configuration is required (see 3.27).

7.3 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PINs to be used for items acquired to this description are created as follows:



Pin Codes:

1 = Size 1A
2 = Size 1B
3 = Size 2A
4 = Size 2B

B = Class B
C = Class C

7.4 Supersession data. This commercial item description replaces Federal Specification KKK-E-2808, dated December 30, 1992.

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7.5 Classification cross reference. Cross reference of classification changes between this commercial item description (see 2.) and the superseded Federal Specification KKK-E-2808 is as follows:

A-A-59250	KKK-E-2808
Size 1A	Size 1
Size 1B	Not designated
Size 2A	Size 2
Size 2B	Not designated
Class B	Class B
Class C	Class C

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Army - MT
Air Force - 84
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 3805-0025)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59250	**2. DOCUMENT DATE (YYMMDD)** 980616

3. DOCUMENT TITLE

EXCAVATOR, MULTIPURPOSE, TRUCK MOUNTED, TWO CABS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER CODE 15E2, 4111 SAN PEDRO ST., PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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